Theoretical minimum: Paper 009

Read the fictional Apartment 50 preface. This is a local practice quiz, not a historical Landau examination or a qualification. **AuthorityEffect: None.** Use the study guide for preparation and the separate instructor key after attempting the questions. Cite a section, definition, input, or artifact for each answer.

The triangle has edges (0,1), (1,2), (2,0), in that order. Potentials live on vertices; edge differences are target minus source. Use integers and exact fractions for new calculations. Historical records retain their representation.

Level 1 — Claim, witness, receipt (Section 1)

1. Rewrite "the system verifies repairs" as one inspectable claim. Name the input, allowed operation, checker, and expected stored state.
2. Distinguish an assumption, witness, receipt, and open obligation with a triangle example. Which can be supplied before execution?
3. A model prints `qed` and an author supplies a correctly hashed output file. What has been established? What check is still needed for the theorem claim?

Level 2 — Input preservation (Section 3)

4. For h=(1/3,1/3,1/3) and f=(0,1/3,2/3), calculate df, h−df, and both cycle sums.
5. Explain why every vertex potential has zero cycle sum in its edge differences. What happens if a constant is added to all vertex potentials?
6. For h=(2,−1,4) and f=(0,2,1), compute h−df. Compare subtraction with replacement of h by df. Which operation changes the declared claim input?

Level 3 — Controls that distinguish cases (Section 4)

7. An LP checker accepts whenever primal and dual objectives match. State the missing conditions. Design separate failures for a primal bound, dual closure, and a feasible but nonmatching objective.
8. Node content changes while its old digest remains. Compare digest recomputation with a check that the digest string has 64 characters.
9. A checker passes a valid example. Does this rule out always-pass behavior? Give a positive/negative pair and name the tested distinction.

Level 4 — Recount the historical bout (Section 4.1)

10. Reconstruct the four cells from six refused proposals (all challenged) and four not-refused proposals (two challenged, two concurred). CHALLENGE is the positive decision; historical refusal is the positive label.
11. Compute each row's challenge fraction, the refused fraction among challenges, and overall label agreement. Give denominators.

12. Why does the caught flag being true in all ten rows not show perfect detection? Explain the later recorded checker replay, and what these trials do not establish about physical ground truth or adversary benefit.

Level 5 — Formal scope and supplied inputs (Sections 5–6)

13. Explain 32 named theorems versus 516 environment declarations. Does inspecting their reports freshly compile Lean?
14. A Lean predicate calls unequal parameter hashes "independent." What does its self-comparison theorem establish, and which independence claim remains open?
15. A toy gate receives caller-supplied zero scores and clean flags and returns `PASS_TO_MASTER`. Name two missing bridges before claiming semantic verification.

Level 6 — Audit the historical gate (Appendix A)

16. F1 starts from (1,1,1), then a lift returns zeros and the gate authorizes. Locate the input change. Is this a potential repair?
17. Explain the paired F2 failures: a counterexample without "DEFECT"/"FLAW" survives, while "No defect found" blocks. How many probes and defect families?
18. F3 hashes a run key and asserts constant witness properties. F4 emits repair prose for a refused, zero-case run. What does digest-chain checking miss?

Level 7 — Corrections and remaining obligations (Appendix A.5)

19. State necessary evidence conditions in the corrected gate for a declared carrier and for no declared carrier. Does an exact zero alone suffice?
20. A candidate and an accepted checker receipt are digest-bound. Separate byte binding, producer authentication, checker correctness, and critique validity.
21. Historical corrected-gate passing counts are cited; this review only inspects source and retained records. Write a defensible result and name the evidence needed to claim a fresh replay.

Level 8 — Reproducible review records (Sections 6–8)

22. Assign review dispositions: (a) displayed arithmetic checks; (b) subtraction is claimed to make the original sum zero; (c) formal/runtime correspondence is absent; (d) a proposed native command was not executed.
23. List essential execution-record fields. Why are sorted keys insufficient for RFC 8785? What if a rate denominator is zero?

24. Repeated potential adjustments leave the residual unchanged. What should the reviewer record? Does a stagnation trigger license changing inputs?

Capstone — Recount, perturb, and report (8 points)

Predict the counts before running `python tools/review_capstone.py` from the package root. Complete one point for each item:

1. Verify the three retained record digests against their bundled provenance.
2. Check ten distinct conditions: two lattices times five interaction values.
3. Reconstruct the four cells and four exact descriptive fractions.
4. Change one refused CHALLENGE to CONCUR in a copy. Predict and recount.
5. Change only the caught flag in a copy. Predict whether cells change.
6. Group all five gate probe records into four defect families.
7. Preserve originals and identify the altered copy as a control, not a new trial.
8. Write source identity, command, observed outcome, scope, and next obligation.

The tool checks custody, condition census, and recount controls. It does not reexecute models, native checkers, or the historical gate. A matching output does not automatically grade the written explanations.

Practice rubric

Each question earns 0–3 points: one for the result, one for calculation or reason, and one for evidence scope or remaining obligation. The 24 questions total 72 points; the capstone adds eight, for 80 possible. Use scores to find material to revisit. No score establishes competence or publication authority.